Course notes

Complex plane

Definition · Image, affix. The plane has a direct orthonormal frame $(O, \vec{u}, \vec{v})$. To the complex number $z = a + ib$ (a, b real) we associate the point $M(a, b)$, called the *image* of z, and the vector $\vec{w}\begin{pmatrix} a \\ b \end{pmatrix}$. We say that z is the *affix* of M and of $\vec{w}$. The plane with this correspondence is called the *complex plane*.

Property · Affixes and operations. If A and B have affixes z_A and z_B, the vector $\overrightarrow{AB}$ has affix $z_B - z_A$ and the midpoint of $[AB]$ has affix $\frac{z_A + z_B}{2}$. The affix of $\vec{w} + \vec{w'}$ is the sum of the affixes, that of $k\vec{w}$ (k real) is k times the affix of $\vec{w}$.

The image of $\overline{z}$ is the reflection of the image of z in the horizontal axis; the image of $-z$ is its reflection in O.

Modulus

Definition · Modulus. The *modulus* of $z = a + ib$ is the nonnegative real number $|z| = \sqrt{a^2 + b^2}$. If M is the image of z, $|z| = OM$. For a real number, the modulus is the absolute value.

Property · Modulus and conjugate. For every complex number z, $|z|^2 = z\overline{z}$, and $|\overline{z}| = |-z| = |z|$. If A and B have affixes z_A and z_B, then $AB = |z_B - z_A|$.

Proof. With $z = a + ib$, $z\overline{z} = a^2 + b^2 = |z|^2$ (chapter 1). The distance AB is the norm of the vector $\overrightarrow{AB}$, whose affix is $z_B - z_A$. □

Property · Modulus of a product, of a power, of a quotient. For all complex numbers z, z' and every natural number n:

$$|zz'| = |z|\,|z'|, \qquad |z^n| = |z|^n, \qquad \text{and, if } z' \neq 0, \quad \left|\frac{1}{z'}\right| = \frac{1}{|z'|}, \quad \left|\frac{z}{z'}\right| = \frac{|z|}{|z'|}.$$

Proof. *Product.* $|zz'|^2 = zz'\,\overline{zz'} = z\overline{z}\,z'\overline{z'} = |z|^2\,|z'|^2$. Moduli are nonnegative, so $|zz'| = |z|\,|z'|$.

Power. By induction: $|z^0| = 1 = |z|^0$, and if $|z^n| = |z|^n$, then $|z^{n+1}| = |z^n|\,|z| = |z|^{n+1}$.

Inverse and quotient. $|z'| \times \left|\frac{1}{z'}\right| = \left|z' \times \frac{1}{z'}\right| = 1$, then $\left|\frac{z}{z'}\right| = |z|\left|\frac{1}{z'}\right|$. □

Property · Triangle inequality, taken for granted here. For all complex numbers z and z', $|z + z'| \leq |z| + |z'|$. The problem of the chapter gives a proof and the case of equality.

Definition · The set $\mathbb{U}$. We write $\mathbb{U}$ for the set of complex numbers of modulus 1. Their images form the circle with centre O and radius 1.

Property · Stability of $\mathbb{U}$. If z and z' belong to $\mathbb{U}$, then $zz' \in \mathbb{U}$ and $\frac{1}{z} \in \mathbb{U}$; moreover $\frac{1}{z} = \overline{z}$.

Proof. $|zz'| = 1 \times 1 = 1$ and $\left|\frac{1}{z}\right| = \frac{1}{1} = 1$. Finally $z\overline{z} = |z|^2 = 1$, so $\overline{z} = \frac{1}{z}$. □

Method · Reading a set of points. Let A and B have affixes a and b, and M have affix z. The equality $|z-a| = r$ $(r > 0)$ means $AM = r$: M describes the circle with centre A and radius r. The equality $|z-a| = |z-b|$ $(a \neq b)$ means $AM = BM$: M describes the perpendicular bisector of $[AB]$.

Argument and trigonometric form

Definition · Argument. Let z be a nonzero complex number with image M. An *argument* of z is any measure, in radians, of the oriented angle $\left(\vec{u}, \overrightarrow{OM}\right)$. We write $\arg z$ for one of them; the others differ from it by a multiple of 2π, which we write $\arg z \equiv \theta \pmod{2\pi}$. The number 0 has no argument.

Property · Trigonometric form. Every nonzero complex number $z = a + ib$ can be written $z = r(\cos\theta + i\sin\theta)$ with $r = |z| > 0$ and $\theta \equiv \arg z \pmod{2\pi}$; then

$$\cos\theta = \frac{a}{r}, \qquad \sin\theta = \frac{b}{r}.$$

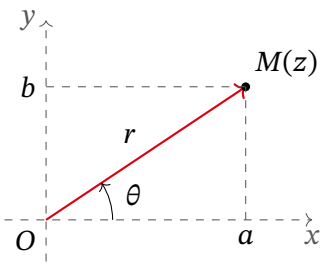
Conversely, if $z = r(\cos\theta + i\sin\theta)$ with $r > 0$, then $|z| = r$ and θ is an argument of z. Two trigonometric forms $r(\cos\theta + i\sin\theta)$ and $r'(\cos\theta' + i\sin\theta')$ are equal if and only if $r = r'$ and $\theta \equiv \theta' \pmod{2\pi}$.

Property · First arguments. For $z \neq 0$: $\arg\overline{z} \equiv -\arg z$ and $\arg(-z) \equiv \arg z + \pi \pmod{2\pi}$. A positive real number has argument 0, a negative real number π; i has argument $\frac{\pi}{2}$.

Method · Finding a trigonometric form. Compute $r = |z|$, then $\frac{a}{r}$ and $\frac{b}{r}$, and recognise the cosine and sine of a standard angle.

Example 1 · Four trigonometric forms. $1 + i\sqrt{3}$: $r = 2$, $\cos\theta = \frac{1}{2}$, $\sin\theta = \frac{\sqrt{3}}{2}$, so $1 + i\sqrt{3} = 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$.
$-1 + i = \sqrt{2}\left(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}\right)$; $-3 = 3(\cos\pi + i\sin\pi)$; $-2i = 2\left(\cos\left(-\frac{\pi}{2}\right) + i\sin\left(-\frac{\pi}{2}\right)\right)$.

Watch out. $-2\left(\cos\frac{\pi}{5} + i\sin\frac{\pi}{5}\right)$ is not a trigonometric form, because $-2 < 0$. We write $-2\left(\cos\frac{\pi}{5} + i\sin\frac{\pi}{5}\right) = 2\left(\cos\frac{6\pi}{5} + i\sin\frac{6\pi}{5}\right)$.



Modulus and argument of $z = a + ib$.

Key point · Key points. $|z| = OM$, $|z_B - z_A| = AB$, $|z|^2 = z\bar{z}$; the modulus of a product is the product of the moduli; $z = r(\cos\theta + i\sin\theta)$ with $r > 0$; an argument is only defined up to a multiple of 2π.